BMC Women's Health

SYSTEMATIC REVIEW Open Access

Beyond the placebo effect: a systematic review and a network meta-analysis of treatment strategies for vulvodynia

Arsenio Spinillo[1,2], Mattia Dominoni[1,2*], Lara Tiranini[1,2], Chiara Cassani[1,2], Rossella Elena Nappi[1,2] and Barbara Gardella[1,2]

Abstract

Background To comprehensively evaluate the comparative and absolute effectiveness of available treatments for vulvodynia on pain and sexual function outcomes through network meta-analysis (NMA).

Methods Systematic searches extracted data from randomized controlled trials (blinded or unblinded) evaluating pharmacological and non-pharmacological treatments in women with vulvodynia, from inception through September 2025. Of 40 studies assessed for eligibility, 34 studies involving 1,893 participants were included in the final analysis. Risk of bias was assessed using the Cochrane Risk of Bias 2 (RoB 2 tool.Confidence in network meta-analysis results was evaluated using the Confidence in Network Meta-analysis (CINeMA) framework.

Results Pain and sexual function outcomes were rescaled to percentage of maximum possible score (POMP, 0–100 scale) to enable pooling across heterogeneous measurement instruments. Both relative treatment effects (comparing interventions to placebo/no treatment) and absolute effects (pre-post changes) were evaluated using Bayesian NMA. Ten treatment categories were analysed: acupuncture, botulinum toxin, topical anaesthetics, topical non-anaesthetics, multimodal/physical therapy, psychological approaches, surgery, tissue-based therapies (photobiomodulation, low-level laser, vaginal transcutaneous electrical nerve stimulation, shockwave therapy), placebo, and no treatment. For pain immediately post-therapy (33 studies, 1,873 women), tissue-based therapies demonstrated superior efficacy compared to placebo in both NMA (POMP difference: -16.1, 95% CI: -24.1, -7.9, mixed evidence, moderate confidence), and pairwise meta-analyses (POMP difference: -20.2, 95% CI: -28.6, -11.7, $I^2 = 77\%$) and ranking highest in surface under the cumulative ranking curve (SUCRA) analysis (85.2%). Pharmacological interventions showed no superiority over placebo. Compared to no treatment, multimodal-physical therapy (POMP: -25.3, 95% CI: -40.8, -9.7, mixed evidence, low confidence), acupuncture (POMP: -25.6, 95% CI: -43.4, -8.1, mixed evidence, moderate confidence), and psychological approaches (POMP -27.4, 95% CI: -44.6, -10.3, indirect evidence, low confidence) all demonstrated moderate-to-high pain reduction. At 3–6 months follow-up (16 studies, 1,106 women), tissue-based treatments remained the only therapy significantly superior to placebo (POMP: -24.5, 95% CI: -42.2, -6.7, mixed evidence, moderate confidence; SUCRA = 77.9%). In absolute effects analysis, placebo produced significant pain reduction (Absolute effect: -13.7, 95% CI: -18.3, -9.3 immediately; -11.7, 95% CI: -17.7, -6.0 at 3–6 months), with

*Correspondence:
Mattia Dominoni
matti.domino@gmail.com

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

confidence intervals overlapping all treatments except tissue-based therapies (Absolute effect: -28, 95% CI: -33.4, -21.7 immediately; -31.6, 95% CI: -43.9, -19.1 at follow-up). Placebo had no effect on sexual function improvement (14 studies, 942 women), (Absolute effect: 2.5, 95% CI: -8., 12.9). At 3–6 months of follow-up (8 studies, 429 women), psychological-based treatments, when added to multimodal treatment (POMP = 9.0, 95% CI = 1.8, 16.7, I^2 = 41.6%), improved modestly sexual functioning.

Conclusions In NMA, tissue-based therapies were the most effective treatment for vulvodynia-associated pain, achieving reductions that clearly exceeded placebo effects with moderate evidence quality. Commonly prescribed pharmacological interventions lack robust evidence of superiority over placebo. Multimodal approaches combined with psychological interventions show promise, particularly for sexual function recovery.

Keywords Vulvodynia, Placebo, Network meta-analysis, Therapy

Introduction

Vulvodynia is a chronic idiopathic pain syndrome characterized by vulvar discomfort, such as burning, stinging, or irritation, often exacerbated by sexual intercourse, tampon insertion, or prolonged sitting [1, 2]. This condition profoundly impairs quality of life, particularly affecting physical functioning, emotional well-being, and sexual relationships, with many women reporting dyspareunia and reduced sexual satisfaction [1, 2]. Epidemiological studies indicate that vulvodynia affects between 8 and 16% of women at some point in their lifetime [1, 2]. Despite its considerable prevalence, fewer than half of affected women seek treatment, and remarkably, only 1.4% receive a correct diagnosis [1]. This diagnostic delay, coupled with the condition's multifactorial aetiology and heterogeneous symptomatic presentation, has contributed to persistent challenges in establishing evidence-based therapeutic guidelines. Therapeutic approaches for vulvodynia are multifaceted, targeting its presumed neuropathic, musculoskeletal, and psychosocial components. Pharmacological options include topical agents like lidocaine, oestrogen creams, or compounded neuromodulators (e.g., gabapentin or amitriptyline) for localized relief, and systemic medications such as tricyclic antidepressants (e.g., desipramine), anticonvulsants (e.g., pregabalin), or serotonin-norepinephrine reuptake inhibitors to address central sensitization [3, 4]. Non-pharmacological interventions encompass pelvic floor physical therapy, biofeedback, cognitive-behavioural therapy (CBT), and emerging modalities like photobiomodulation or internet-based programs combining education and mindfulness [3, 4]. A critical limitation of the existing evidence base is that the majority of randomized controlled trials in vulvodynia have enrolled relatively small numbers of participants, substantially limiting statistical power to detect clinically meaningful treatment effects [4, 5]. The evaluation of treatment efficacy in vulvodynia is additionally complicated by substantial placebo effects documented across multiple randomized controlled trials [4, 5]. Systematic reviews indicate that placebo responses can exceed 40–50% in pain reduction, with no pharmacological intervention consistently outperforming placebo across studies [5]. Although pairwise meta-analyses have attempted to synthesize evidence on vulvodynia treatments, such as those showing small effects for nonpharmacological therapies, standard meta-analytic investigations are constrained by reliance on direct comparisons, limiting integration of the diverse interventions and often precluded by heterogeneity [5, 6]. Network meta-analysis (NMA) addresses many of these limitations by allowing indirect comparisons between treatments that have not been directly compared in head-to-head trials, incorporating more comprehensive data, and providing a broader perspective on the therapeutic options [7]. Furthermore, network meta-analysis offers methodological advantages including increased statistical power, the ability to rank treatments, and more precise effect estimates [8]. Critically, for conditions like vulvodynia where placebo effects are substantial, Bayesian network meta-analysis enables the estimation of both relative treatment effects (comparing active interventions against each other or placebo) and absolute effects (quantifying the magnitude of change from baseline for each intervention). These absolute effect estimates are essential for clinical decision-making, as they allow clinicians and patients to understand not only which treatment is superior in comparative terms, but also the expected magnitude of benefit that can be anticipated from each therapeutic option including placebo [9]. The present network meta-analysis was conducted to comprehensively evaluate the comparative and absolute effectiveness of available treatments for vulvodynia on pain and sexual function outcomes.

Materials and methods

The protocol for this network meta-analysis (NMA) was registered in PROSPERO (n. CRD420251140836). We included both blinded and unblinded randomized studies that comparatively assessed the impact of pharmacological and nonpharmacological treatments on pain intensity and sexual functioning in women with vulvodynia. Following the PICO framework (Population,

intervention, comparator, outcome), the population consisted of women diagnosed with vulvodynia (primary or secondary, provoked, spontaneous, mixed, or undefined) according to standard methods [10]. Specifically, the diagnosis included women with vulvar pain of unknown aetiology lasting more than three months. No age restriction was applied. Studies including women of any age (reproductive, perimenopausal, or postmenopausal) were eligible provided they met the vulvodynia diagnostic criteria. Interventions comprised the therapeutic procedures evaluated in each study; comparators (controls) included placebo, sham treatment, no treatment, or other procedures; and outcomes included measurements of pain and sexual functioning before and after treatment. Given the wide variety of treatments used to manage vulvodynia symptoms, we initially categorized treatments and comparators as follows: 1. Acupuncture. 2. Botulinum toxin injections 3. Topical anaesthetic treatment (lidocaine). 4. Topical non-anaesthetic treatment (oestrogen, nifedipine, steroids, diazepam, emollient creams). 5. Multimodal/physical therapy (a comprehensive approach including physical therapy, pelvic floor muscle exercises, massage and cognitive behavioural therapy involving active behavioural desensitization) 6. Psychological approaches (psychotherapy, mindfulness, and cognitive therapies focused solely on acceptance) that were typically used as adjunctive treatments in women already receiving topical or multimodal treatments. 7. Surgical treatment. 8. Tissue-based therapies (vaginal transcutaneous electrical nerve stimulation, photobiomodulation, shockwave therapy, low-level laser therapy). 9. Systemic treatment (gabapentin, amytriptyline, desipramine, others) 10. Placebo (true pharmacological placebo or sham treatment for nonpharmacological approaches). 11. No treatment (waitlist or no additional treatment provided). Pharmacological treatments were analysed separately or as a whole category. Treatment categories were created based on several considerations. First, individual agents within categories (e.g., different topical non-anaesthetics or systemic medications) were each evaluated in only 1–3 studies with small sample sizes, making separate analysis statistically underpowered and resulting in very wide, imprecise confidence intervals. Second, while we acknowledge that treatments within categories have different specific mechanisms of action (e.g., topical oestrogen addresses hormonal factors while topical diazepam targets muscle hypertonicity), they share a common route of administration and clinical positioning as local therapies, reflecting pragmatic treatment choices in clinical practice. Third, creating overly granular treatment nodes would have resulted in a sparse, disconnected network that would prevent indirect comparisons and defeat the purpose of network meta-analysis. Fourth, this approach follows established precedent in network meta-analyses of complex interventions where treatments are grouped into clinically meaningful categories to balance statistical feasibility with interpretability. Finally, we conducted sensitivity analyses examining pharmacological treatments both individually and combined to assess whether grouping influenced conclusions about their efficacy. Finally, we adhered to the standard NMA principle for "Add-on" designs: when a background therapy was present in both the intervention and control arms, the node assignment was determined by the distinct intervention being tested [7–9]. We assumed that the relative effect of the distinct intervention (e.g., Diazepam) compared to its control was consistent, regardless of the background therapy (e.g., Physical Therapy).

Search methods and eligibility criteria

The search strategy was published in PROSPERO (https://www.crd.york.ac.uk/PROSPERO/view/CRD42025114083 6) and reported in Suppl. 1. In brief, the main search terms included "vulvodynia," "vestibulodynia," "vulvar pain," or "vulvar vestibulitis" combined with "randomized." Databases searched were Embase, Embase via Ovid, MEDLINE, PubMed, Science Citation Index, Scopus, and the Cochrane Central Register of Controlled Trials. Pain and sexual function were measured before treatment initiation, immediately post-treatment and 3–6 months thereafter.

Study selection

Three reviewers (MD, BG, LT) independently screened abstracts of selected studies and extracted data from full-text articles. The extracted data were tabulated and compared for consistency. Only treatments evaluated in at least two studies were included in the analysis. The study selection process is reported in a PRISMA diagram (Fig. 1). Of the 40 studies assessed for eligibility [11–50], three studies [46, 47, 50] were excluded due to unavailable data, one study was a randomized investigation of two different doses of the same treatment (no comparator) [49], and two studies involved unique interventions that could not form viable treatment nodes: vestibulectomy [45] (the only eligible surgical study, precluding a surgical node with the required minimum of two studies, and transcranial direct current stimulation [48] leaving 34 studies for analysis (Table 1). Three included studies used a crossover design [12, 19, 21]. To avoid unit-of-analysis errors and eliminate potential carry-over effects, we extracted only data from the first period of these trials, before participants crossed over to the alternate treatment. This approach, recommended by the Cochrane Handbook (Sect. 23.2.4), effectively treats crossover trials as parallel-group trials and avoids bias from period effects or insufficient washout [51]. Consequently, we applied the standard Cochrane RoB 2 tool rather than

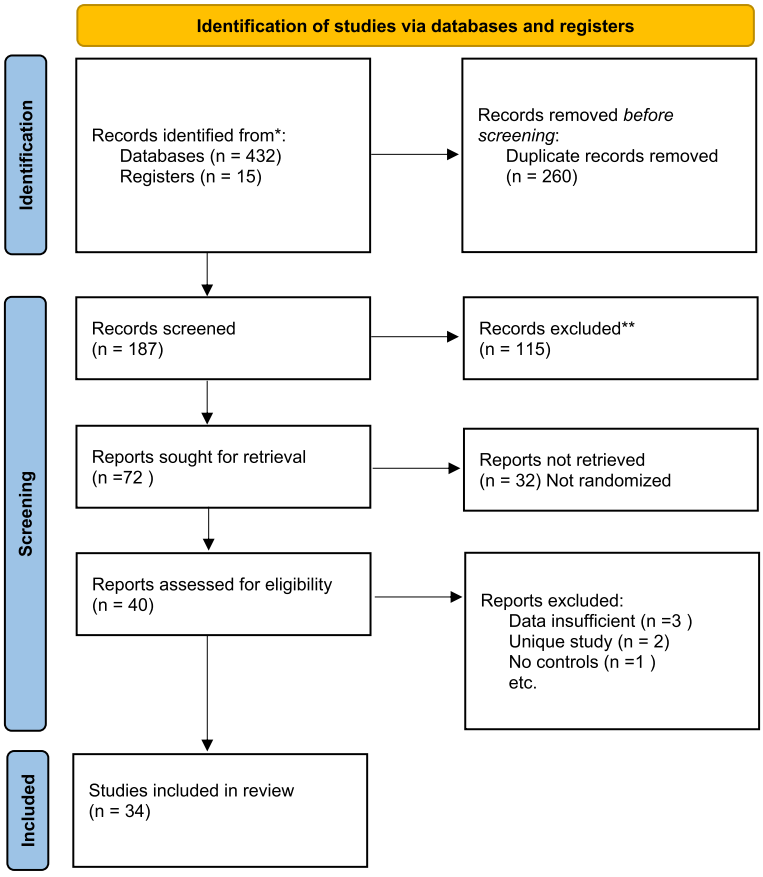


Fig. 1 PRISMA flow chart of research strategy

the crossover-specific tool, as we analysed only the pre-crossover phase. One included study [17] compared two different psychological interventions (mindfulness-based cognitive therapy vs. cognitive behavioural therapy) without a placebo or no-treatment control arm. We included this study because these represent genuinely distinct therapeutic modalities (cognitive vs psychological) with different theoretical frameworks, techniques, and mechanisms of action, allowing network meta-analysis to indirectly estimate their comparative effectiveness through connections to other studies. Finally, categories included acupuncture [43], botulinum toxin [20, 24, 29], psychological [14, 15, 17, 28, 34, 35, 40], multimodal [23, 26, 36, 44] and tissue-based therapies including photobiomodulation [11], low-level laser therapy [33], shockwave therapy [25, 31] and TENS [37, 39, 41].

Assessment of risk of bias

The risk of bias (RoB) in individual studies was assessed using the revised Cochrane Risk of Bias tool for randomized trials (RoB 2) by three reviewers (MD, BG, LT) [51], with oversight by a senior author (AS). The graphical and analytical classification of RoB for each study is reported in Supplementary Material 1.

Synthesis of results

Pain intensity in the included studies was measured using various methods and at different sites or situations (e.g., Q-tip test, tampon test, pain during intercourse). The Visual numeric scale was used in 4 studies [14, 15, 17, 28], the McGill Pain Questionnaire in 2 studies [19, 35], and other methods (Likert, Friedrich, etc.) in the remaining 6 studies [12, 22, 26, 27, 34, 40]. When multiple pain evaluation sites were reported, we prioritized measures in the following order: cotton swab testing on the vulvar vestibule, pressure threshold testing, and reported pain during intercourse. Sexual functioning was measured using the Female Sexual Function Index (FSFI) in 13 of the 14 studies reporting this outcome. To evaluate treatment effects on pain intensity and sexual function, we rescaled all scores to a 0–100 scale and expressed severity of pain as a percentage of the maximum possible score (POMP) [52, 53]. This unitless and intuitive measure avoids standardized mean differences, which are less intuitive and more statistically problematic for evaluating relative and absolute treatment effects on psychometric measures [53, 54]. Meta-analyzed data included the absolute difference in pre-post POMP scores (POMP differences) and their corresponding standard deviations,

Table 1 Characteristics and main results of included randomised controlled trials

Author, Year Design; Country	N	Age (yr)	Intervention	Comparator	Treatment Category	Pain Scale Baseline POMP (%)	Main Outcome Results
1. Tissue-based therapies							
Antonio et al., 2025 RCT; Canada [11]	30	27.0	Photobiomodulation (PBM)/Laser 3 sessions/week × 4 weeks (12 sessions total)	Sham PBM	Tissue-based	Tampon pain test[1] (0–10); VPAQ POMP: 42.5%	Laser significantly reduced tampon pain and vulvar pressure pain vs sham (between-group d = 0.60–0.87). No difference in satisfaction between groups
Gruenwald et al., 2021 RCT; Israel [25]	32	27.5	Extracorporeal shockwave therapy (ESWT) 1 session/week × 6 weeks	Sham ESWT (same schedule)	Tissue-based	VAS[2] (0–10); CST POMP: 85.0%	ESWT significantly reduced pain vs sham at 1, 4, and 12 weeks post-treatment (VAS $p < 0.01$; CST $p < 0.01$). Pain reduction > 30% consistently in ESWT group
Hurt et al., 2020 RCT; Czech Republic [31]	62	39.5	Extracorporeal shockwave therapy (ESWT) 1 session/week × 12 weeks	Sham ESWT (same schedule)	Tissue-based	VAS[2] (0–10) POMP: 61.5%	ESWT significantly reduced pain vs sham (POMP Δ: − 35% vs 0%). Significant between-group difference
Lev-Sagie et al., 2017 RCT; Israel [33]	34	26.0	Low-level laser therapy (LLLT) 2 sessions/week × 6 weeks (12 sessions total)	Sham LLLT (same schedule)	Tissue-based	VAS[2] (0–100); Q-tip test POMP: 88.0%	78% of LLLT patients reported clinical improvement vs 44% of sham ($p = 0.042$); objective measures (Q-tip, VAS) not significantly different between groups
Murina et al., 2008 RCT; Italy [37]	40	27.0	Vaginal TENS 3 sessions/week, ~ 8 weeks (mean 26.7 sessions)	Sham TENS (same schedule)	Tissue-based	VAS[2] (0–10); SF-MPQ; Marinoff POMP: 65.0%	TENS significantly improved pain (VAS $p = 0.004$; SF-MPQ $p = 0.001$) and sexual function (FSFI, Marinoff) vs sham
Tissue-based/Pharmacological – systemic							
Murina et al., 2013 RCT; Italy [39]	20	32.0	PEA 400 mg + vaginal TENS (home protocol) Twice-daily PEA; mean 26.7 TENS sessions	Placebo + TENS (same home protocol)	Tissue-based/Pharmacological – systemic	VAS[2] (0–10); Marinoff; CPT POMP: 60.0%	PEA + TENS significantly reduced pain vs placebo + TENS (VAS $p < 0.01$; dyspareunia $p < 0.01$); both groups improved. Greater benefit with shorter disease duration
Tissue-based/Pharmacological – topical							
Murina et al., 2018 RCT; Italy [38]	42	28.5	Vaginal diazepam 5 mg + TENS Treatment over 60 days	Placebo suppositories + TENS	Tissue-based/Pharmacological – topical	VAS[2] (0–10); Marinoff; surface EMG POMP: 73.0%	Both groups significantly reduced pain ($p = 0.01$). Diazepam group significantly superior for pelvic floor relaxation and C-fiber thresholds ($p < 0.05$). No significant between-group VAS difference
2. Botulinum toxin							
Diomande et al., 2019 RCT; France [20]	32	27.5	Botulinum toxin A (50 or 100 units) Single vestibular injection	Saline injection (placebo)	Botulinum toxin	VAS[2] (0–10) POMP: 70.0%	Both groups improved; no significant between-group difference in pain VAS at follow-up
Goldstein et al., 2025 RCT; USA [24]	60	29.5	Botulinum toxin A (100 units) Single vestibular injection	Saline injection (placebo)	Botulinum toxin	VAS[2] (0–10) POMP: 80.0%	No significant between-group difference in primary pain outcome (VAS)

Table 1 (continued)

Author, Year Design; Country	N	Age (yr)	Intervention	Comparator	Treatment Category	Pain Scale Baseline POMP (%)	Main Outcome Results
Haraldson et al., 2020 RCT; Sweden [29]	88	26.5	Botulinum toxin A (50 units) Single vestibular injection	Saline injection (placebo)	Botulinum toxin	VAS[2] (0–100); vaginal manometer POMP: 67.5%	No significant between-group difference in primary outcome (dyspareunia VAS). Botulinum significantly reduced tampon pain (secondary, – 11 VAS units) and pelvic floor tone
Petersen et al., 2009 RCT; Denmark [41]	64	30.0	Botulinum toxin A (20 units) Single vestibular injection	Saline injection (placebo)	Botulinum toxin	VAS[2] (0–10) POMP: 70.5%	No significant between-group difference in primary outcome (dyspareunia). Both groups improved similarly (POMP Δ: – 10% each)
Pharmacological – systemic							
Bachman et al., 2019 RCT; USA [12]	64	36.0	Oral gabapentin ER 1200–3000 mg/day 6-week titration and treatment period	Placebo	Pharmacological – systemic	Tampon pain test[1] (0–6); NRS POMP: 43.5%	Gabapentin significantly improved overall sexual function vs placebo ($p = 0.008$); subgroup benefit in women with high baseline pelvic pain (FSFI, $p = 0.02$)
Brown et al., 2009 RCT; USA [19]	43	50.0	Amitriptyline 10 mg/day or topical steroid Once daily × 8 weeks	Self-management program	Pharmacological – systemic	MPQ[4] (0–78) POMP: 29.0%	No significant between-group differences on pain scores (MPQ); all three groups improved. Low baseline severity (POMP 29%)
Brown et al., 2018 DBCT crossover; USA [18]	89	36.5	Oral gabapentin ER 1200–3000 mg/day 16-week crossover (pre-crossover phase analysed)	Placebo	Pharmacological – systemic	Q-tip test[5] (0–10); NRS POMP: 63.5%	Gabapentin did not significantly reduce tampon test pain ($p = 0.07$), daily pain (p = 0.36), or intercourse pain ($p = 0.76$) vs placebo
Pharmacological – systemic/Local anesthetic							
Foster et al., 2010 RCT crossover; USA [22]	99	32.5	Oral desipramine (up to 150 mg/day) or topical lidocaine 5% (nightly); 12-week titration	Placebo (oral tablet + placebo cream)	Pharmacological – systemic/ Local anesthetic	Vulvalgesiometer[6] (0–28 g/cm^2)POMP: 67.0%	No significant between-group differences; all groups improved substantially. Lidocaine and desipramine provided similar pain relief to placebo
Pharmacological – systemic							
Grigol Bardin et al., 2020 RCT; Brazil [26]	57	29.0	Amitriptyline 25 mg/day Daily oral × 8 weeks	Pelvic floor physical therapy (weekly sessions × 8 weeks)	Pharmacological – systemic	Friedrich criteria[7] (0–15) POMP: 67.5%	Physical therapy significantly superior to amitriptyline for pain reduction (POMP Δ: – 32% vs – 18%); physical therapy favored for all secondary outcomes
Pharmacological – topical							
Bazin et al., 2011 RCT; Canada [13]	61	30.5	Conjugated estrogen vaginal cream 1.875 g Daily application × 6 weeks	Placebo	Pharmacological – topical	VAS[2] (0–10); dyspareunia POMP: 73.0%	Both groups significantly reduced dyspareunia (p < 0.01); no significant between-group difference (p = 0.5). Estrogen reduced vestibular inflammation
Bornstein et al., 2010 RCT; Israel [16]	30	24.5	Nifedipine 0.2% or 0.4% topical 4 × daily application × 6 weeks	Placebo	Pharmacological – topical	VAS[2] (0–100); Q-tip test POMP: 88.5%	All groups improved at 3 months; nifedipine 0.2% reduced speculum pain immediately post-treatment; no significant between-group difference in dyspareunia at 3 months

Table 1 (continued)

Author, Year Design; Country	N	Age (yr)	Intervention	Comparator	Treatment Category	Pain Scale Baseline POMP (%)	Main Outcome Results
Donders et al., 2012 RCT; Belgium [21]	30	26.5	Fibroblast lysate (cromolyn) 4% cream Twice-daily application × 3 months	Placebo cream (twice daily)	Pharmacological – topical	VAS[2] (1–10) POMP: 72.0%	Cromolyn cream significantly reduced pain vs placebo (POMP Δ: – 17% vs – 2%); statistically significant between-group difference
Langlais et al., 2017 RCT; Canada [32]	20	24.0	Conjugated equine estrogen cream Daily application × 8 weeks	Placebo cream (daily × 8 weeks)	Pharmacological – topical	VAS[2] (0–10); FSFI; McGill POMP: 61.0%	Estrogen significantly improved all secondary outcomes (McGill, FSFI, sexual satisfaction) vs placebo ($p < 0.05$); no significant between-group difference on VAS ($p = 0.29$)
Nyirjesy et al., 2001 RCT; USA [40]	26	26.5	Cromolyn sodium 4% cream Frequency 3 daily; 3-month treatment period	Placebo cream	Pharmacological – topical	VAS[2] (0–15) POMP: 60.0%	No significant between-group difference; both groups improved. ~ 38–46% of patients achieved ≥ 50% symptom reduction regardless of group
5. Physiotherapy/Multimodal							
Hess Engström et al., 2022 RCT; Sweden [30]	88	24.0	Internet-based multimodal physiotherapy Self-administered program × 12 weeks	No treatment (waitlist)	Physiotherapy/Multimodal	VAS[2] (0–10) POMP: 69.0%	Internet-based physiotherapy significantly reduced pain vs no treatment (POMP Δ: – 36% vs – 8%)
Merlino et al., 2024 RCT; Italy [44]	20	22.6	Pelvic floor PT + intravaginal diazepam Weekly sessions; duration 2 weeks	Pelvic floor PT alone	Physiotherapy/Multimodal	VAS[2] (0–10); Marinoff scale POMP: 58.0%	Combined therapy significantly superior to PT alone: VAS 5.8 → 2.8 ($p < 0.001$); Marinoff 2.3 → 1.4 ($p = 0.009$); significant improvement in pelvic floor biometrics
Morin et al., 2021 RCT; Canada [36]	212	22.0	Multimodal pelvic floor physiotherapy Weekly sessions × 10 weeks	Topical lidocaine 5% (nightly × 10 weeks)	Physiotherapy/Multimodal	NRS[3] (0–10); FSFI, FSDS, PGIC POMP: 73.0%	Multimodal PT significantly superior to lidocaine for intercourse pain (POMP Δ: – 46% vs – 28%, $p < 0.001$); 79% vs 39% reported significant improvement (PGIC, $p < 0.001$). Maintained at 6-month follow-up
6. Psychological/Behavioral							
Bergeron et al., 2016 RCT; Canada [14]	97	27.0	CBT + topical corticosteroid cream Weekly sessions × 13 weeks	Topical corticosteroids alone	Psychological/Behavioral	NRS[3] (0–10); FSFI POMP: 74.5%	No significant between-group FSFI difference at post-treatment or 24-week follow-up; CBT favored on composite sexual outcomes when controlling for baseline imbalance
Bergeron et al., 2021 RCT; Canada [15]	108	26.0	Cognitive-behavioral couple therapy Weekly sessions × 12 weeks	Overnight topical lidocaine	Psychological/Behavioral	NRS[3] (0–10); FSFI, FSDS POMP: 71.5%	No significant between-group FSFI difference. CBT group showed significantly greater reduction in sexual distress (FSDS) at post-treatment ($p < 0.001$)
Brotto et al., 2019 RCT; Canada [17]	130	32.0	CBT or Mindfulness-based CBT (MBCT) 8 weekly sessions of 2 h 25 min	Mindfulness-based CBT (MBCT)	Psychological/Behavioral	NRS[3] (0–10); vulvalgesiometer POMP: 66.3%	Both CBT and MBCT significantly reduced vulvar pain (~ 68% and 58% reduction, respectively); 59–60% reported moderate-to-excellent improvement at 6-month follow-up. No significant between-group difference

Table 1 (continued)

Author, Year Design; Country	N	Age (yr)	Intervention	Comparator	Treatment Category	Pain Scale Baseline POMP (%)	Main Outcome Results
Goldfinger et al., 2016 RCT; Canada [23]	20	26.0	Cognitive-behavioral therapy 8 sessions of 1.5 h over 24 weeks	Pelvic floor physical therapy (8 sessions over 24 weeks)	Psychological/Behavioral	Q-tip test[5] (0–10) POMP: 51.5%	Both CBT and physical therapy reduced pain similarly (POMP Δ ≈ – 25%). No significant between-group difference
Guillet et al., 2019 RCT; France [27]	31	32.0	Mindfulness-based pain group therapy 8 weekly group sessions	Pain education support group (8 weekly sessions)	Psychological/Behavioral	Likert scale[8] (0–10) POMP: 45.0%	Both approaches reduced pain; mindfulness showed greater reduction (POMP Δ: – 19% vs – 12%). No formal between-group significance test reported
Halbedl et al., 2025 RCT; Austria [28]	33	29.0	Mindfulness-based cognitive therapy (MBCT) Weekly sessions × 10 weeks	Pelvic floor physiotherapy (weekly sessions × 10 weeks)	Psychological/Behavioral	NRS[3] (0–5) POMP: 89.0%	Mindfulness significantly superior to physiotherapy for pain reduction (POMP Δ: – 26% vs – 8%). Significant between-group difference
Maathz et al., 2023 RCT; Sweden [34]	44	27.0	Online Acceptance & Commitment Therapy (ACT) Self-guided online program × 12 weeks	Waitlist control	Psychological/Behavioral	FSFI pain domain[9] (0–6) POMP: 42.0%	Online ACT had no significant effect on pain or sexual function; significant improvement in quality of life, and moderate effect on anxiety
Masheb et al., 2009 RCT; USA [35]	50	34.5	Cognitive-behavioral therapy (CBT) 10 weekly individual sessions	Supportive psychotherapy (SPT) (10 weekly sessions)	Psychological/Behavioral	McGill PQ[4] (0–78); speculum/cotton-swab rating POMP: 34.0%	CBT significantly superior for sexual function ($p = 0.034$) and physician-rated pain ($p = 0.014$). Treatment effects maintained at 1-year follow-up
7. Acupuncture							
Rubin et al., 2019 RCT; USA [42]	19	29.0	Acupuncture 10 sessions over 5 weeks (2 ×/week)	Sham acupuncture (same schedule)	Acupuncture	VAS[2] (0–100) POMP: 65.0%	Both groups improved substantially; no significant between-group difference in pain. High placebo response observed
Shlaeger et al., 2015 RCT; USA [43]	36	35.0	Acupuncture 10 sessions over 5 weeks (2 ×/week)	Waitlist control	Acupuncture	VAS[2] (0–10); Q-tip; FSFI POMP: 56.5%	Acupuncture significantly reduced pain vs no treatment (POMP Δ: – 29% vs – 6%). Significant improvement in FSFI and Q-tip scores

POMP (Percent of Maximum Possible score) = [(baseline pain score – scale minimum)/(scale maximum – scale minimum)] × 100. Higher values indicate greater pain severity. Baseline POMP represents the study-level mean across all arms

POMP Percentage of maximum possible score. *CBT* Cognitive-behavioral therapy, *CPT* Current perception threshold, *DBCT* Double-blind crossover trial, *EMG* Electromyography, *ER* Extended-release, *ESWT* Extracorporeal shockwave therapy, *FSDS* Female Sexual Distress Scale, *FSFI* Female Sexual Function Index, *LLLT* Low-level laser therapy, *MPQ* McGill Pain Questionnaire, *MBCT* Mindfulness-based cognitive therapy, *NR* Not reported, *NRS* Numerical Rating Scale, *PBM* Photobiomodulation, *PEA* Palmitoylethanolamide, *PGIC* Patient Global Impression of Change, *POMP* Percent of Maximum Possible score, *PT* Physical therapy, *RCT* Randomised controlled trial, *SF-MPQ* Short-Form McGill Pain Questionnaire, *TENS* Transcutaneous electrical nerve stimulation, *VAS* Visual Analogue Scale, *VPAQ* Vulvar Pain Assessment Questionnaire

[1]Tampon pain test: standardised tampon insertion rated on a 0–6 or 0–10 numeric scale (0 = no pain)

[2]VAS (Visual Analogue Scale): 100-mm or 10-cm horizontal line; anchors 0 = no pain, 100 (or 10) = worst imaginable pain

[3]NRS (Numerical Rating Scale): 11-point scale (0 = no pain, 10 = worst imaginable pain)

[4]McGill Pain Questionnaire (MPQ): multidimensional pain measure, total pain rating index 0–78 (higher = more pain)

[5]Q-tip test: pain rated on a 0–10 NRS during standardised q-tip palpation of the vulvar vestibule

[6]Vulvalgesiometer: pressure-pain threshold device; scored 0–28 g/cm^2 (lower threshold = greater sensitivity)

[7]Modified Friedrich criteria: composite 0–15 scale rating vestibular tenderness, erythema, and introital entry pain

[8]Likert scale (0–10): global pain severity, adapted for provoked vestibulodynia

[9]FSFI pain domain: subscale of the Female Sexual Function Index assessing pain during intercourse (0–6; lower = more pain)

computed according to the Cochrane Handbook for Systematic Reviews of Interventions, accounting for the correlation between pre- and post-treatment scores [55]. In the main analysis, standard deviations of POMP differences were calculated assuming a high correlation coefficient between pre- and post-treatment measurements ($r = 0.8$) [53, 55]. For the evaluation of relative treatment effects, a sensitivity analysis was performed using POMP differences and standard deviations calculated assuming moderate ($r = 0.5$) and low ($r = 0.3$) correlations between pre-post measurements. Differences in pre-post POMP scores between treatments were used to compute relative effects [53]. We applied the IMMPACT recommendations [9] to POMP scores to facilitate a clinical, as well as a statistical, interpretation of our findings. According to these recommendations, a ~ 30% reduction from baseline is considered a moderately important change, and a 10% reduction is considered a minimally important change. For the overall median baseline POMP of 67 in this study (IQR = 15.4), these correspond to absolute reductions of approximately 20 and 7 POMP points, respectively. Both thresholds are used throughout the results to contextualise effect sizes. Reducing pain after treatments was computed as increasingly negative POMP scores whereas improvement of sexual function as increasingly positive scores. Main analysis was performed using a standard random-effects Bayesian network meta-analysis and was conducted using GEMTC (General Mixed Treatment Comparisons) and BUGS (Bayesian Inference Using Gibbs Sampling) routines in R (version 4.5.2 The R Foundation for Statistical Computing 2025) to evaluate the relative effects of treatments compared to placebo and no treatment [56, 57]. Network plots were generated using ggplot2 package in R. GEMTC and BUGS routines were also used for pairwise comparisons, nodesplit analysis, ranking analysis, and meta-regression to evaluate the role of potential moderators of effects. We assessed potential small-study effects by including study sample size as a covariate in a Bayesian random-effects network meta-regression model, implemented in GEMTC [57, 58]. Ranking analysis was performed by evaluating the surface under the cumulative ranking curve (SUCRA). To evaluate model inconsistency and heterogeneity, we compared the deviation of the inconsistency model with that of the NMA model [56, 57]. Differences in the Deviance Information Criterion (DIC) between models with and without covariates in meta-regression analysis were used to assess the influence of moderators on relative effects, with DIC differences > 3 considered significant [58]. Previous vulvodynia trials [6] suggest that placebo has a significant effect on pain, and absolute pre-post changes in pain scores are critical for evaluating treatment efficacy clinically [9]. Therefore, we also evaluated the absolute effects of treatments, including placebo, as differences in pre-post POMP scores using a Bayesian arm-based model with the PCNETMETA routine in R [59]. Confidence in the NMA results was assessed using the CINeMA web-based program [60], which simultaneously and semi-automatically evaluates six domains: within-study bias, reporting bias, indirectness, imprecision, heterogeneity, and incoherence. Based on the results for these domains in each analysis, confidence was graded from high to very low [60]. CINeMA assessments were performed for all NMA relative-effect comparisons involving pain-reduction (immediately post-therapy and 3–6 months follow-up) and for sexual function outcomes where NMA was feasible. For the 3–6 month sexual function outcome, where the network was disconnected, CINeMA assessment was not applicable.

Results

Study selection and network characteristics

The final analysis included 34 studies and 1893 subjects with vulvodynia (Table 1). Twenty-one studies included only women with provoked vulvodynia [11, 12, 14, 15, 18–20, 22–24, 28, 31, 34, 35, 37, 38, 41–44] while the remaining 13 included women with mixed or undefined forms [13, 16, 17, 21, 25–27, 29, 30, 32, 33, 36, 39, 40]. The overall median age of enrolled women was 28 years (SD = 6.2). Studies were conducted in the USA (10 studies) [12, 18, 19, 22, 24, 27, 35, 40, 42, 43] Canada (8 studies) [11, 13–15, 17, 23, 32, 36], Europe (12 studies) [20, 21, 28–31, 34, 37–39, 41, 44] Israel (3 studies) [16, 25, 33], and Brazil (one study) [26].

Risk of bias and certainty of evidence

Risk of bias was assessed across 34 RCTs using the RoB 2 tool (Suppl 1). Overall, four studies (12%) were rated at low risk [11, 21, 24, 25], 26 (76%) at some concerns [12–16, 18–20, 22, 23, 27, 29, 31–44], and four (12%) at high risk [17, 26, 28, 30]. The most consistently problematic domain was selection of the reported result (D5), which raised concerns in 27 studies, primarily owing to absent pre-registration. The randomisation process (D1) was rated at high risk in three studies [17, 30, 36] and at some concerns in four [22, 23, 31, 42]. Missing outcome data (D3) and outcome measurement (D4) were rated at low risk in all or nearly all studies. Deviations from intended interventions (D2) raised some concerns in 11 studies, largely reflecting open-label designs inherent to behavioural and physical interventions.

Certainty of NMA estimates was evaluated using CINeMA (Suppl. 2–4. Reporting bias was at low risk across all comparisons. For pain immediately post-therapy, the placebo vs tissue-based comparison (5 direct studies: [11, 25, 31, 33, 37]) achieved moderate certainty, downgraded for heterogeneity; local non-anaesthetics vs tissue-based ([38]) and systemic vs tissue-based ([39]) were both rated

low, downgraded for within-study bias and imprecision or heterogeneity. Among comparisons vs no treatment, acupuncture ([43]) was rated moderate (heterogeneity), multimodal/physical therapy ([30, 34]) was rated low (within-study bias, heterogeneity), and no treatment vs systemic therapies (indirect network evidence through [19, 26]) was rated very low, additionally penalised for imprecision and incoherence. For pain at 3–6 months, placebo vs tissue-based (4 direct studies: [11, 25, 31, 37]) was moderate (heterogeneity), while all remaining comparisons, botulinum, local anaesthetics, local non-anaesthetics, multimodal/physical, and psychological vs tissue-based, were low based on indirect evidence, downgraded for indirectness and imprecision. For sexual functioning post-therapy, placebo vs tissue-based (3 direct studies: [11, 25, 37]) and all indirect comparisons were rated low, due to within-study bias, heterogeneity, and imprecision. No comparison reached high certainty across any outcome.

Pain immediately post-therapy

A total of 1,873 women reported pain severity before and immediately after therapy in 33 studies across 10 treatment types (Fig. 2A). The overall median POMP score before starting therapy was 67 (IQR = 15.4) suggesting that initial pain was moderate. For a population with this severity of initial pain, a 30% reduction (moderate effect) corresponds to an absolute reduction of approximately 20 POMP points. A 10% reduction from baseline, often considered a minimally important change, corresponds to an absolute reduction of approximately 7 POMP points in this analysis. This translation from relative percent reduction to absolute pre-post percent reduction of pain allows for a direct clinical interpretation of our effect estimates. Tissue-based therapies were associated with a significant decrease in pain scores compared to placebo in both network (POMP = −16.1, 95% CI = −24.2, −7.9, mixed evidence, moderate confidence) and pairwise (POMP = −20.2, 95% CI = −28.6, −11.7, I^2 = 77%, moderate effect) meta-analyses (Fig. 2C) (Suppl.2) In pairwise analysis, multimodal treatment was superior to no treatment in reducing pain (POMP = −27.3, 95% CI = −39.1, −14.8, I^2 = 14.4%, moderate effect). None of the local or systemic pharmacological therapies evaluated separately (Fig. 2C) or as a whole (POMP = −3.68, 95% CI = −9.23, 1.92) (Suppl. 2) were better than placebo in reducing pain. Tissue-based therapies also ranked highest in the SUCRA analysis (SUCRA = 85.2%) (FI G.2B). Node-split analysis and the overall test of treatment-by-design interaction (p = 0.35) suggested that the model was consistent (Suppl.2). Multimodal-physical (POMP = −25.3, 95% CI = −40.8, −9.7, mixed evidence, low confidence), acupuncture (POMP = −25.6, 95% CI = −43.4, −8.1, mixed evidence, moderate confidence), psychological (POMP = −27.4, 95% CI = −44.7, −10.3, indirect evidence, low confidence), and tissue-based approaches

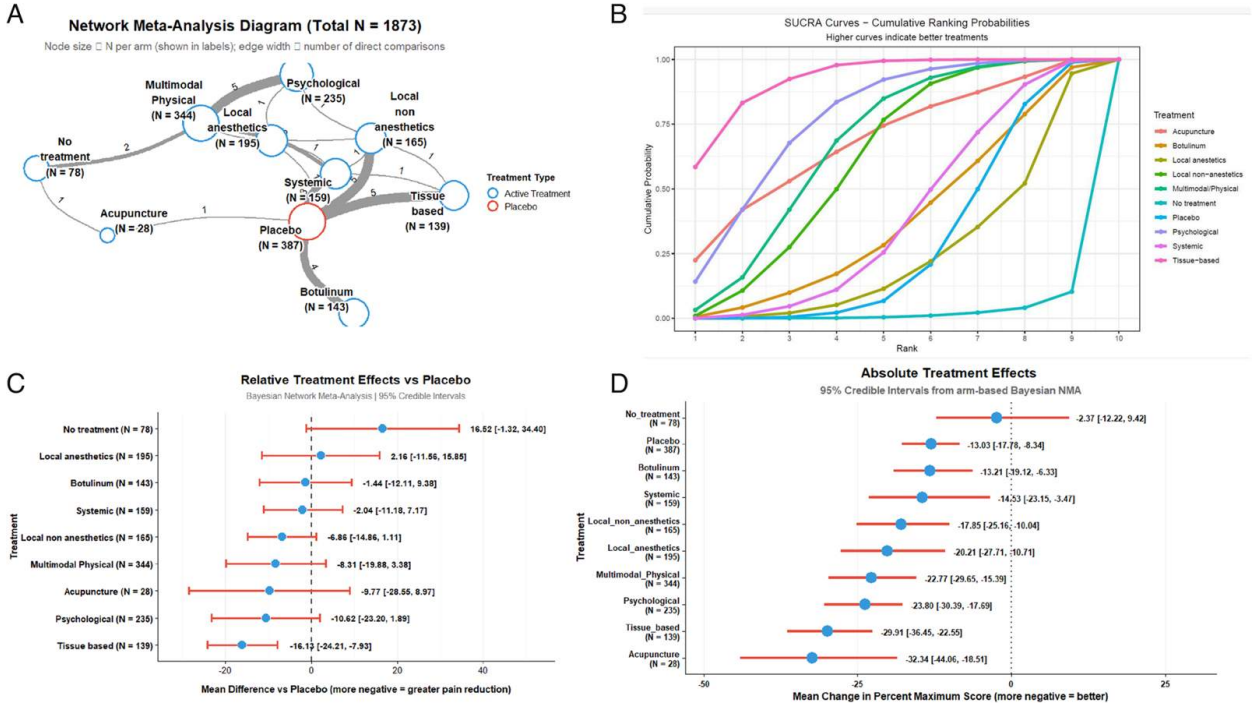


Fig. 2 Network characteristics (**A**), relative effects (**B**), surface under the cumulative ranking curve (SUCRA)(**C**) and absolute effects (**D**) of different treatments in vulvodynia as obtained by Bayesian network meta-analysis. (Pain reduction Immediately post therapy). (Effects = percentages of maximum possible score)

(POMP = −30.9, 95% CI = −49.5, −12.1, indirect evidence, low confidence) were all associated with a moderate-to-high reduction in pain compared to no treatment (Suppl.2). Placebo was associated with a modest but significant absolute reduction in pain score (Absolute effect = −13, 95% CI = −17.8, −8.3), with a 95% confidence interval that overlapped with all other therapies except tissue-based treatments (Absolute effect = −29.9, 95% CI = −36.4, −22.5) (Fig. 2D). Potential moderators evaluated in meta-regression included initial POMP score (initial pain intensity), mean age of subjects, type of vulvodynia (provoked vs. others), RoB values (high vs. moderate/low), percent of dropouts in the study and sample sizes. The initial DIC value of the Bayesian model without covariates was 130.5. The addition of initial POMP scores (DIC = 130.9), mean age (DIC = 130.8), type of vulvodynia (DIC = 131), RoB values (DIC = 131), percent of dropouts (DIC = 131.3), sample size (DIC = 130.6) and year of publication (DIC = 130.9) did not significantly alter the relationship between treatments and relative effects. Sensitivity analyses with POMP variation computed at moderate (r = 0.5) and low (r = 0.3) correlations between pre-post measurements confirmed that tissue-based therapies were still associated with a significant reduction in pain (Suppl. 2).

Pain at 3–6 months of follow-up

Follow-up data on pain 3–6 months after the end of treatment were reported in 16 studies involving 1106 women (Fig. 3A). Only one study had a no-treatment control arm [30]; to simplify the model, this single arm was included in the placebo category. This merging may have modestly influenced the placebo effect estimate at follow-up, as no-treatment and placebo differ in in level of therapeutical contact and patient expectation; this is noted as a minor analytical limitation. In both NMA (POMP = −24.6, 95% CI = −42.3, −6.8, mixed evidence, moderate confidence) and pairwise (POMP = −25.2, 95% CI = −34.0, −16, I^2 = 75.3%, moderate effect) meta-analyses, tissue-based treatments were the only therapies associated with reduced pain at 3–6 months post-therapy compared to placebo. (Fig. 3C) (Suppl.3) Both the global test for a random treatment-by-design interaction (p = 0.83) and node-split analysis (Suppl. 3) were non-significant, suggesting model consistency. Tissue-based treatments ranked highest in the SUCRA analysis (SUCRA = 77.9%) (FI G.3B). In the evaluation of absolute effects, placebo was still associated with a significant reduction in pain score at 3–6 months (Absolute effect = −11.7, 95% CI = −17.7, −6.0), with a 95% confidence interval that overlapped with all treatments except tissue-based therapies (Absolute effect = −31.6, 95% CI = −43.9, −19.2). (Fig. 3D) Sensitivity analyses at moderate and low correlations between pre-post scores confirmed these results (Suppl.3). DIC

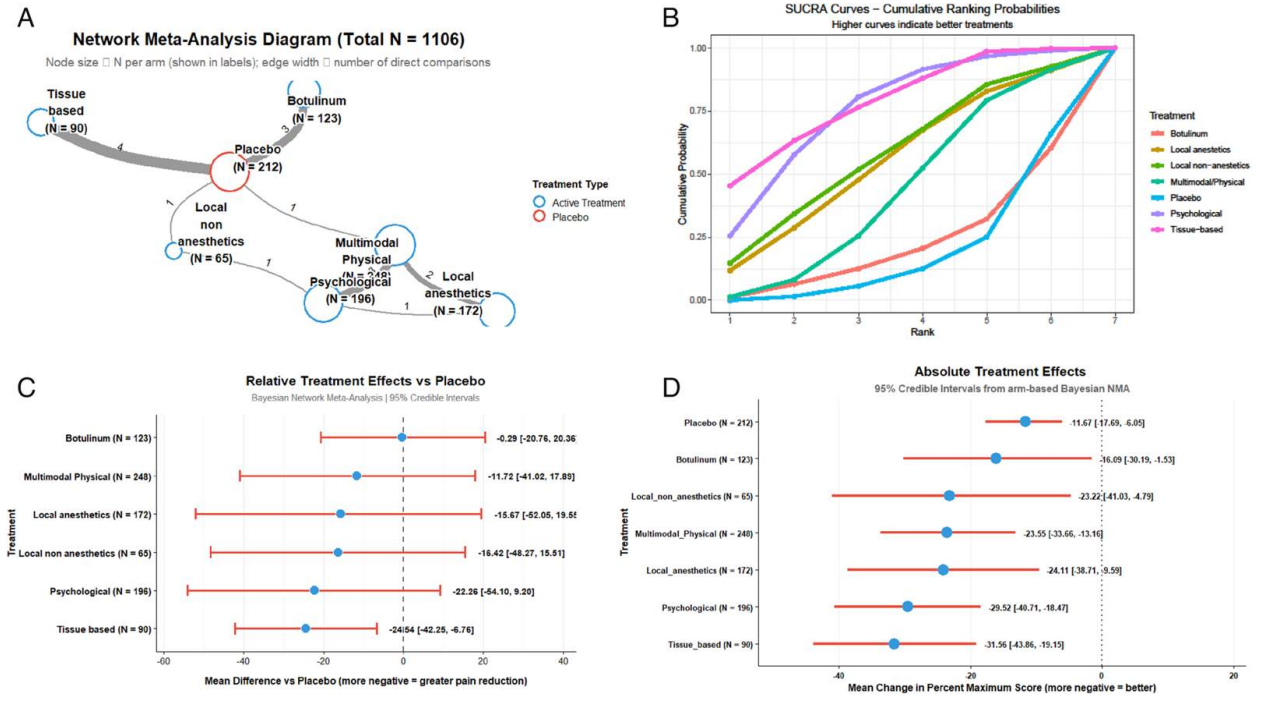


Fig. 3 Network characteristics (**A**), relative effects (**B**), surface under the cumulative ranking curve SUCRA) (**C**) and absolute effects (**D**) of different treatments in vulvodyniaas obtained by Bayesian network meta-analysis. (Pain reduction after 3–6 months of follow-up). (Effects = percentages of maximum possible score)

for the model without covariates was 63.6. Corresponding DIC values for regression models with covariate age (DIC = 64), type of vulvodynia (DIC = 63.9), initial severity of pain (DIC = 63.5), small study effect (DIC = 62.8), ROB (DIC = 63.7), percent of dropouts (DIC = 62.1) and year of publication (DIC = 64.1) suggested that the model was consistent.

Sexual function modification immediately and 3–6 months after treatment

The network for the immediate effects on sexual function included 14 studies and 942 women (Fig. 4A). In pairwise meta-analysis, tissue-based treatment improved sexual function compared to placebo by increasing POMP scores (13.5, 95% CI = 3.6, 23.3, I^2 = 70.6%, low effect) (Suppl. 4); in the NMA, this effect was of borderline significance (POMP = 13.4, 95% CI = −0.3, 27, mixed evidence, low confidence) (Fig. 4C). Tissue-based treatment ranked highest in the SUCRA analysis (71.8%) (FI G.4B). Evaluation of absolute effects showed that, in contrast to pain, placebo had no effect on the improvement of sexual function (Absolute effect = 2.5, 95% CI = −8, 12.9). In the analysis of absolute effect, psychological and multimodal/physical approaches increased modestly sexual functioning (Fig. 4D). The model was consistent as demonstrated by the random-treatment-by design interaction (p = 0.43) and node-split analysis (Suppl. 4). Sensitivity analysis at different correlation coefficients between pre-post scores confirmed the results (Suppl.4). The effect of treatments on sexual functioning 3–6 months after treatment was evaluated in 8 studies involving 429 women. The network was disconnected, making NMA impossible. In pairwise meta-analysis (Fig. 5), psychological-based treatments, when added to multimodal treatment alone (POMP = 9.0, 95% CI = 1.6, 16.5, I^2 = 42.9%, low effect), and tissue-based treatments compared to placebo (POMP = 12.2, 95% CI = 5.0, 19, I^2 = 34.4%, low effect) improved sexual functioning at 3–6 months post-therapy.

Discussion

In this network meta-analysis evaluating the comparative and absolute effectiveness of various treatments for vulvodynia, tissue-based therapies (photobiomodulation, shockwave therapy, TENS, low-level laser therapy) emerged as the most effective intervention for reducing pain intensity both immediately post-therapy and at 3–6 months follow-up when compared to placebo, with moderate confidence in the evidence. These interventions also ranked highest in probabilistic analysis and, critically, exhibited absolute effect sizes with confidence intervals that did not overlap with placebo effects, indicating a magnitude of benefit clearly exceeding non-specific treatment factors. These therapies were grouped as a single category because they share a common mechanism of delivering non-invasive physical energy, such as light, electrical stimulation, or acoustic waves, to

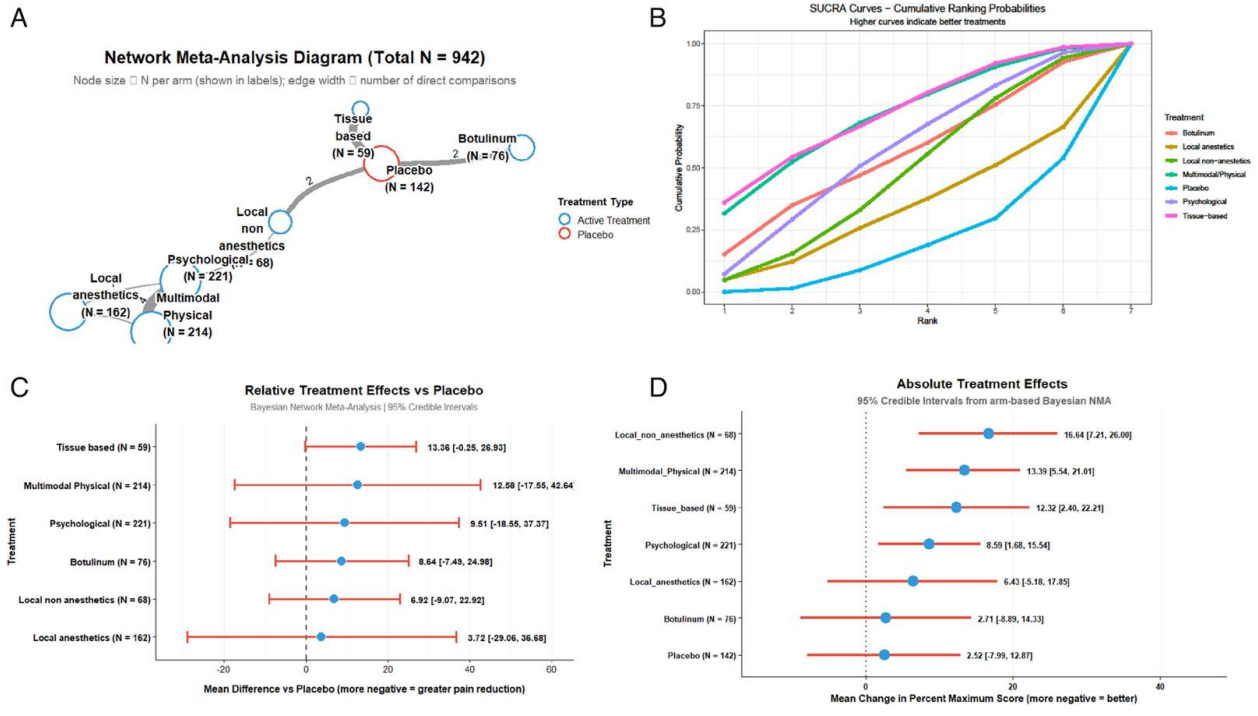


Fig. 4 Network characteristics (**A**), relative effects (**B**), surface under the cumulative ranking curve (SUCRA) (**C**) and absolute effects (**D**) of different treatments in vulvodynia as obtained by Bayesian network meta-analysis. (Sexual functioning improvement immediately after therapy). (Effects = percentages of maximum possible score)

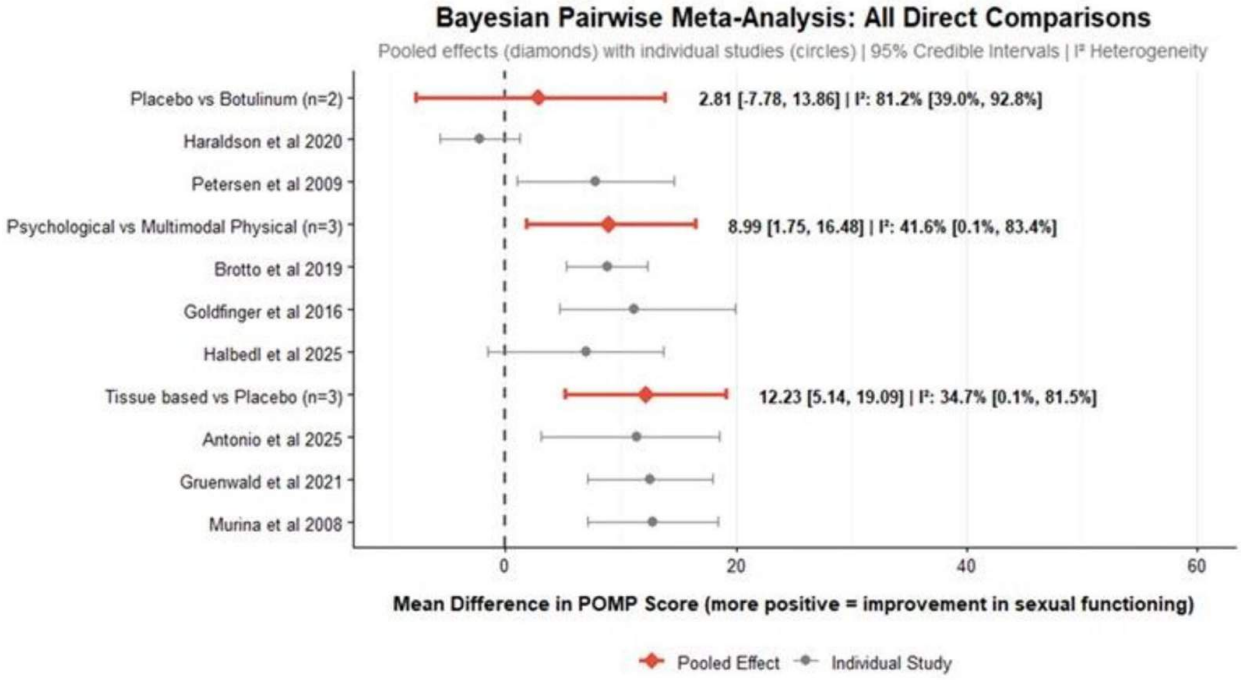


Fig. 5 Pairwise meta-analysis on sexual functioning improvement after 3–6 months of therapy. (Effects = percentages of maximum possible score)

modulate local vulvar tissue, enabling a unified analysis of their collective efficacy despite differences in specific modalities [3, 5, 10]. Results should be interpreted at the treatment-category level and, given the grouping of heterogeneous interventions within network nodes, should be regarded as exploratory and hypothesis-generating rather than definitive. The grouping strategy, while methodologically necessary to maintain a connected network and achieve sufficient statistical power, limits inference about any single agent within a category and represents a central methodological limitation of this study. In particular, the heterogeneity of agents within the tissue-based category,spanning distinct physical mechanisms such as photobiomodulation, acoustic shockwave, low-level laser, and electrical stimulation,means that the observed category-level effect estimate cannot be extrapolated to any individual modality without further dedicated evidence.

Individual modalities within the tissue-based category are each supported by only 1–3 studies; category-level effect estimates should therefore not be interpreted as equivalently strong evidence for each specific intervention. The mechanisms underlying tissue-based therapies likely involve neuromodulation, improved tissue perfusion, reduced inflammation, and enhanced cellular repair through stimulation of mitochondrial ATP production and release of endogenous analgesics [10, 11, 31, 37]. Contrary to widespread clinical use [61], pharmacological interventions, whether topical anaesthetics, topical non-anaesthetic treatments, or systemic neuromodulators, failed to demonstrate statistically significant superiority over placebo in pain reduction. This finding persisted across both individual drug categories and when pharmacological treatments were analysed collectively. Other non-pharmacological approaches, such as acupuncture demonstrated moderate pain reduction vs. no treatment supported by moderate-certainty evidence. In contrast, multimodal and psychological approaches showed similar effect sizes but with low-certainty evidence, indicating that these results should be interpreted more cautiously. Regarding sexual function outcomes, tissue-based treatments showed borderline significant improvement immediately post-treatment, while psychological interventions added to multimodal treatment demonstrated benefit at medium-term follow-up. A striking finding of this NMA was the substantial placebo response observed across studies. Placebo alone was associated with a mean absolute pain reduction of 13.7% (95% CI: −18.3, −9.3) immediately post-therapy and 11.7% (95% CI: −17.8, −6.0) at 3–6 months but without significant effect on sexual functioning. This robust placebo effect highlights the role of psychological factors and patient expectations in pain management, suggesting that future interventions should incorporate strategies to harness these non-specific effects. Additionally, the modest improvements in sexual function underscore the need for integrated approaches that address not only pain but also relational and emotional dimensions, potentially through combined therapies. From a practical standpoint, the accessibility and cost-effectiveness of tissue-based therapies warrant further investigation, as they may vary by healthcare

setting and could influence adoption in clinical practice. Patient preferences, including tolerance for invasive procedures versus non-invasive options, should also guide treatment selection to optimize adherence and outcomes.

Strengths and limitations

A major strength of this study is its methodological comprehensiveness, integrating both relative and absolute treatment effects through a Bayesian NMA. This approach allowed simultaneous comparison across diverse treatment classes, increasing statistical power, facilitating treatment ranking, and addressing limitations inherent in traditional pairwise meta-analyses that are often constrained by heterogeneity and sparse direct evidence [7, 8]. The use of percentage of maximum scores, rather than standardized mean differences, permitted pooling of heterogeneous scales while preserving clinical interpretability [52–54]. Furthermore, by modelling absolute pre–post changes, we quantified the therapeutic magnitude of both active treatments and placebo, enabling an assessment of efficacy beyond relative comparisons. The use of sensitivity analyses to evaluate the role of pre-post measurements, the use of meta-regression for potential moderators of effects, and the Confidence in Network Meta-Analysis framework further strengthened the validity of the results. Although sample size was not a significant predictor in the meta-regression, small studies may have limited power to detect subtle effects and increased the imprecision of the estimates, as indicated by the CINeMA assessments. Heterogeneity in outcome measures (e.g., varying pain assessment methods like VAS or NRS) and vulvodynia subtypes (provoked versus mixed) could have introduced bias, although meta-regression analyses suggested these factors did not significantly moderate effects. The magnitude of placebo effects varied across studies, complicating the interpretation of relative treatment effects. Factors such as patient expectations, the therapeutic context, and the nature of the placebo intervention (e.g., sham procedures vs. oral placebos) may have contributed to this variability. In the 3–6 month follow-up analysis, a single no-treatment arm was merged with the placebo node for network connectivity. Since no-treatment and placebo differ in the level of therapeutic contact and patient expectation, this merging may have modestly inflated or deflated the placebo effect estimate at follow-up. This represents a minor analytical limitation of the follow-up model. Finally, the majority of studies were conducted in high-income countries, which may limit the generalizability of the findings to low- and middle-income settings where access to specialized treatments (e.g., tissue-based therapies) may be restricted. Despite this, most included trials received a rating of 'some concerns' or 'high risk' under RoB 2, primarily due to the inherent difficulty of blinding participants to non-pharmacological interventions, limiting certainty of effect estimates. Furthermore, the included studies were predominantly conducted in premenopausal women (overall median age 28 years, SD = 6.2); findings may not generalise to postmenopausal women, in whom hormonal contributions to vulvodynia may differ substantially. A further limitation concerns the core assumptions of network meta-analysis. Transitivity, the requirement that effect modifiers be similarly distributed across trials,may not be fully satisfied given differences in vulvodynia subtype, patient age, and the nature of comparator arms across included studies. Similarly, grouping agents with distinct mechanisms of action within the same node (e.g., topical oestrogen and topical diazepam) may affect the interpretability of indirect comparisons. Comparisons relying exclusively on indirect evidence should therefore be interpreted with particular caution. Importantly, however, the global tests for inconsistency were non-significant across all outcomes, and sensitivity analyses confirmed the stability of the main findings, providing some reassurance regarding the internal validity of the network.

Comparison with existing literature

Although, to our knowledge, this is the first network meta-analysis evaluating treatments for vulvodynia, in the past years several standard meta-analyses both on pharmacological and non-pharmacological treatment of vulvodynia have been published [3–6, 63–66]. Starzec-Proserpio et al. [5], using standardized mean differences as outcome measures, conducted a meta-analysis on non-pharmacological therapies for chronic pelvic pain (including vulvodynia subsets) and reported small to moderate effects for physical therapy and psychological interventions on pain. Our findings corroborate the benefit of multimodal approaches compared to no treatment, though we found less consistent evidence of superiority over placebo when examining vulvodynia-specific populations. Differences in the population studied (chronic pelvic pain vs. exclusive vulvodynia) and methods (traditional meta-analysis vs. NMA, relative effects vs. relative + absolute effects) used could explain the discrepancies. According to Bohm-Starke et al. [4], who conducted a systematic review limited to provoked vulvodynia, our study acknowledges the scarcity of high-quality trials and the heterogeneity of interventions. They concluded that multimodal physiotherapy showed the most promise, though their synthesis remained qualitative due to inconsistent outcome measures. The placebo effect found in this NMA is consistent with previous reports in vulvodynia and other chronic pain conditions, where placebo responses can exceed 40–50% [6]. The inclusion of absolute effect estimates in this NMA is particularly noteworthy, as it allows clinicians and patients

to contextualize the magnitude of benefit that can be expected from each treatment, including placebo. Our analysis matches the result of a previous classical meta-analysis showing that placebo-associated pain reductions are clinically meaningful and that overlap with most active pharmacological treatments [6]. This NMA and previous data strengthen confidence in the conclusion that commonly prescribed pharmacological interventions for vulvodynia lack robust evidence of superiority over placebo [6, 62, 65]. Physiotherapy and rehabilitative techniques have been evaluated in two additional meta-analyses using both a qualitative and a classic quantitative approach showing a beneficial effect on pain reduction although, small sample sizes and heterogeneity reduced the certainty of the results [63–66]. Finally, a recent meta-analysis on psychotherapy [64] found significant improvements in vulvar pain, sexual function, and psychological adjustment (e.g., pain acceptance), with low certainty, aligning with our low-confidence benefits for psychological interventions over no treatment; however, limited data (only two studies) emphasize the need for rigorous trials.

Clinical implications

Given their comparatively superior efficacy in reducing pain at the category level, tissue-based therapies may deserve greater attention in the clinical evaluation of treatment options for vulvodynia-associated pain. However, this recommendation must be framed within the limitations of the available evidence: findings reflect category-level estimates derived from grouping modalities with distinct mechanisms (photobiomodulation, low-level laser, TENS, shockwave therapy), each individually supported by only 1–3 small trials. Based on moderate-certainty evidence for the category as a whole compared to placebo, these approaches warrant consideration for patients with prominent neuropathic pain characteristics, [11, 25, 31, 33, 37] but definitive prioritisation of any individual modality,or over other established interventions,requires confirmation from larger, well-designed, head-to-head trials. The exploratory nature of these findings should be communicated explicitly to patients when discussing treatment options. Commonly prescribed pharmacological interventions, including topical anaesthetics and systemic neuromodulators, lack robust evidence of superiority over placebo in reducing pain, challenging current practice patterns and highlighting the need for re-evaluation of first-line therapeutic approaches. Improvements in sexual function in this NMA were generally modest across treatments studied. Tissue-based therapies showed borderline significance in improving sexual function compared to placebo, while psychological interventions combined with multimodal therapy demonstrated more sustained benefits at 3–6 months. These findings suggest that while pain reduction is a critical first step, addressing the psychological and relational aspects of vulvodynia is essential for restoring sexual health [15, 35]. In particular, multimodal approaches combining physical therapy with psychological interventions show promise for medium-term sexual function improvement, supporting the conceptualization of vulvodynia as a biopsychosocial condition requiring comprehensive rather than single-modality treatment [5]. While multimodal and psychological approaches did not consistently demonstrate superiority over placebo for pain outcomes, their effects on sexual function and their benefit compared to no treatment support their inclusion in comprehensive care models. Finally, the significant placebo effect on pain outcome underscores the importance of therapeutic relationship, patient expectation management, and psychological factors in treatment outcomes.

Conclusions

This network meta-analysis provides the most comprehensive comparative effectiveness evaluation of vulvodynia treatments to date, using Bayesian dual-analysis to disentangle relative and absolute treatment effects in the context of substantial placebo responses. At the category level, tissue-based therapies demonstrated the most consistent pain reduction, with effects clearly exceeding placebo under moderate-certainty evidence and confirmed across both pairwise and network analyses and at multiple timepoints. While the grouping of heterogeneous interventions within nodes renders these findings exploratory with respect to individual modalities, the convergence of evidence across multiple analytical approaches supports their overall direction. Pharmacological interventions lacked evidence of superiority over placebo. Multimodal approaches combined with psychological interventions show promise for sexual function recovery, though on the basis of low-certainty evidence. Future research should prioritise well-powered trials of individual modalities, direct head-to-head comparisons, and longer follow-up in diverse populations.

Abbreviations

NMA: Network meta-analysis
PICO: Population, intervention, comparator, outcome
ROB: Risk of bias
VAS: Visual Analog Scale
NRS: Numeric Rating Scale

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12905-026-04469-0.

Additional file 1.

Additional file 2.

Additional file 3.

Additional file 4.

Additional file 5.

Authors' contributions
A.S.: wrote the manuscript and performed statistical analysis M.D.: data curation L.T.: data curation C.C: data curation R.E.N: validation B.G: edited the manuscript. All authors reviewed the manuscript.

Funding
No funding to declare.

Data availability
All data generated or analysed during this study are included in this published article and its supplementary information files (Supplementary 5).

Declarations

Ethics approval and consent to participate
Not applicable, Considering that this paper is a systematic review and meta-analysis.

Consent for publication
Not applicable, considering that this study did not include clinical details, clinical images or other identifiable participant data.

Competing interests
The authors declare no competing interests.

Author details
[1]Department of Clinical, Surgical, Diagnostic and Paediatric Sciences, University of Pavia, Strada Nuova, 65, Pavia 27100, Italy
[2]Department of Obstetrics and Gynaecology, Fondazione IRCCS Policlinico San Matteo, Viale Camillo Golgi 19, Pavia 27100, Italy

Received: 6 February 2026 / Accepted: 8 April 2026
Published online: 14 April 2026

References

1. Reed BD, Harlow SD, Sen A, Legocki LJ, Edwards RM, Arato N, et al. Prevalence and demographic characteristics of vulvodynia in a population-based sample. Am J Obstet Gynecol. 2012;206(2):170.e1-9.
2. Xie Y, Shi L, Xiong X, Wu E, Veasley C, Dade C. Economic burden and quality of life of vulvodynia in the United States. Curr Med Res Opin. 2012;28(4):601–8.
3. Goldstein AT, Pukall CF, Brown C, Bergeron S, Stein A, Kellogg-Spadt S. Vulvodynia: Assessment and Treatment. J Sex Med. 2016;13(4):572–90.
4. Bohm-Starke N, Ramsay KW, Lytsy P, Nordgren B, Sjöberg I, Moberg K, et al. Treatment of Provoked Vulvodynia: A Systematic Review. J Sex Med. 2022;19(5):789–808. https://doi.org/10.1016/j.jsxm.2022.02.008.
5. Starzec-Proserpio M, Frawley H, Bø K, Morin M. Effectiveness of nonpharmacological conservative therapies for chronic pelvic pain in women: a systematic review and meta-analysis. Am J Obstet Gynecol. 2025;232(1):42–71. https://doi.org/10.1016/j.ajog.2024.08.006.
6. Miranda Varella Pereira G, Soriano Marcolino M, Silveira Nogueira Reis Z, Vale de Castro Monteiro M. A systematic review of drug treatment of vulvodynia: evidence of a strong placebo effect. BJOG. 2018;125(10):1216–24. https://doi.org/10.1111/1471-0528.15223.
7. Salanti G. Indirect and mixed-treatment comparison, network, or multiple-treatments meta-analysis: many names, many benefits, many concerns for the next generation evidence synthesis tool. Res Synth Methods. 2012;3(2):80–97.
8. Florez ID, De La Cruz-Mena JE, Veroniki AA. Network meta-analysis: a powerful tool for clinicians, decision-makers, and methodologists. J Clin Epidemiol. 2024;176:111537. https://doi.org/10.1016/j.jclinepi.2024.111537.
9. Dworkin RH, Turk DC, Farrar JT, Haythornthwaite JA, Jensen MP, Katz NP, et al. Core outcome measures for chronic pain clinical trials: IMMPACT recommendations. Pain. 2005;113(1–2):9–19. https://doi.org/10.1016/j.pain.2004.09.012.
10. Bergeron S, Reed BD, Wesselmann U, Bohm-Starke N. Vulvodynia. Nat Rev Dis Primers. 2020;6(1):36. https://doi.org/10.1038/s41572-020-0164-2.
11. Antonio F, Pukall C, McLean L. Photobiomodulation therapy for the treatment of vulvar pain among those with provoked vestibulodynia: a randomized controlled trial. J Sex Med. 2025;22(4):579–87. https://doi.org/10.1093/jsxmed/qdaf011.
12. Bachmann GA, Brown CS, Phillips NA, Rawlinson LA, Yu X, Wood R, et al. Effect of gabapentin on sexual function in vulvodynia: a randomized, placebo-controlled trial. Am J Obstet Gynecol. 2019;220(1):89.e1-89.e8. https://doi.org/10.1016/j.ajog.2018.10.021.
13. Bazin S, Lefebvre J, Fortier M, Brisson J, Brouillette F, Bujold E, et al. Évaluation d'une crème œstrogénique vaginale dans le traitement de la vestibulodynie provoquée : essai randomisé à double insu [Evaluation of an estrogen vaginal cream for the treatment of dyspareunia: a double-blind randomized trial]. J Obstet Gynaecol Can. 2011;33(8):838–43. https://doi.org/10.1016/S1701-2163(16)34987-8.
14. Bergeron S, Khalifé S, Dupuis MJ, McDuff P. A randomized clinical trial comparing group cognitive-behavioral therapy and a topical steroid for women with dyspareunia. J Consult Clin Psychol. 2016;84(3):259–68. https://doi.org/10.1037/ccp0000072.
15. Bergeron S, Vaillancourt-Morel MP, Corsini-Munt S, Steben M, Delisle I, Mayrand MH, et al. Cognitive-behavioral couple therapy versus lidocaine for provoked vestibulodynia: a randomized clinical trial. J Consult Clin Psychol. 2021;89(4):316–26. https://doi.org/10.1037/ccp0000631.
16. Bornstein J, Tuma R, Farajun Y, Azran A, Zarfati D. Topical nifedipine for the treatment of localized provoked vulvodynia: a placebo-controlled study. J Pain. 2010;11(12):1403–9. https://doi.org/10.1016/j.jpain.2010.03.016.
17. Brotto LA, Bergeron S, Zdaniuk B, Driscoll M, Grabovac A, Sadownik LA, et al. A comparison of mindfulness-based cognitive therapy vs cognitive behavioral therapy for the treatment of provoked vestibulodynia in a hospital clinic setting. J Sex Med. 2019;16(6):909–23. https://doi.org/10.1016/j.jsxm.2019.04.002.
18. Brown CS, Bachmann GA, Wan J, Foster DC, Gabapentin (GABA) Study Group. Gabapentin for the treatment of vulvodynia: a randomized controlled trial. Obstet Gynecol. 2018;131(6):1000–7. https://doi.org/10.1097/AOG.0000000000002617.
19. Brown CS, Wan J, Bachmann G, Rosen R. Self-management, amitriptyline, and amitripyline plus triamcinolone in the management of vulvodynia. J Womens Health. 2009;18(2):163–9. https://doi.org/10.1089/jwh.2007.0676.
20. Diomande I, Gabriel N, Kashiwagi M, Ghisu GP, Welter J, Fink D, et al. Subcutaneous botulinum toxin type A injections for provoked vestibulodynia: a randomized placebo-controlled trial and exploratory subanalysis. Arch Gynecol Obstet. 2019;299(4):993–1000. https://doi.org/10.1007/s00404-019-05043-w.
21. Donders GG, Bellen G. Cream with cutaneous fibroblast lysate for the treatment of provoked vestibulodynia: a double-blind randomized placebo-controlled crossover study. J Low Genit Tract Dis. 2012;16(4):427–36. https://doi.org/10.1097/LGT.0b013e31825a2274.
22. Foster DC, Kotok MB, Huang LS, Watts A, Oakes D, Howard FM, et al. Oral desipramine and topical lidocaine for vulvodynia: a randomized controlled trial. Obstet Gynecol. 2010;116(3):583–93. https://doi.org/10.1097/AOG.0b013e3181e9e0ab.
23. Goldfinger C, Pukall CF, Thibault-Gagnon S, McLean L, Chamberlain S. Effectiveness of cognitive-behavioral therapy and physical therapy for provoked vestibulodynia: a randomized pilot study. J Sex Med. 2016;13(1):88–94. https://doi.org/10.1016/j.jsxm.2015.12.003.
24. Goldstein A, Rubin R, Dahir M, Goldstein I, Faught BM, Bohm-Starke N, et al. Phase 2 randomized study of abobotulinumtoxinA in patients with provoked vestibulodynia: dose-finding results. J Sex Med. 2025;22(4):588–96. https://doi.org/10.1093/jsxmed/qdaf022.
25. Gruenwald I, Gutzeit O, Petruseva A, Gartman I, Lowenstein L. Low-intensity shockwave for treatment of vestibulodynia: a randomized controlled therapy trial. J Sex Med. 2021;18(2):347–52. https://doi.org/10.1016/j.jsxm.2020.11.006.
26. Bardin MG, Giraldo PC, Martinho N. Pelvic Floor Biometric Changes Assessed by 4D Translabial Ultrassound in Women With Vulvodynia Submitted to Physical Therapy: A Pilot Study of a Randomized Controlled Trial. J Sex Med. 2020;17(11):2236–46. https://doi.org/10.1016/j.jsxm.2020.07.020.
27. Guillet AD, Cirino NH, Hart KD, Leclair CM. Mindfulness-Based Group Cognitive Behavior Therapy for Provoked Localized Vulvodynia: A Randomized

Controlled Trial. J Low Genit Tract Dis. 2019;23(2):170–5. https://doi.org/10.1097/LGT.0000000000000456. (**PMID: 30688760**).
28. Halbedl H, Pfabigan DM, Ebhardt I, Trutnovsky G. Mindfulness-based body scan training in multimodal physiotherapy for vulvodynia - a randomized controlled feasibility study. J Psychosom Obstet Gynaecol. 2025;46(1):2531057. https://doi.org/10.1080/0167482X.2025.2531057.
29. Haraldson P, Mühlrad H, Heddini U, Nilsson K, Bohm-Starke N. Botulinum toxin A as a treatment for provoked vestibulodynia: a randomized controlled trial. Obstet Gynecol. 2020;136(3):524–32. https://doi.org/10.1097/AOG.0000000000004008.
30. Hess Engström A, Bohm-Starke N, Kullinger M, Hesselman S, Högberg U, Buhrman M, et al. Internet-based treatment for vulvodynia (EMBLA) - a randomized controlled study. J Sex Med. 2022;19(2):319–30. https://doi.org/10.1016/j.jsxm.2021.11.019.
31. Hurt K, Zahalka F, Halaska M, Rakovicova I, Krajcova A. Extracorporeal shock wave therapy for treatment of vulvodynia: a prospective, randomized, double-blind, placebo-controlled study. Eur J Phys Rehabil Med. 2020;56(2):169–74. https://doi.org/10.23736/S1973-9087.20.05903-1.
32. Langlais EL, Lefebvre J, Maheux-Lacroix S, Bujold E, Fortier M, Bouchard C. Treatment of secondary vestibulodynia with conjugated estrogen cream: a pilot, double-blind, randomized placebo-controlled trial. J Obstet Gynaecol Can. 2017;39(6):453–8. https://doi.org/10.1016/j.jogc.2016.10.011.
33. Lev-Sagie A, Kopitman A, Brzezinski A. Low-level laser therapy for the treatment of provoked vestibulodynia-a randomized, placebo-controlled pilot trial. J Sex Med. 2017;14(11):1403–11. https://doi.org/10.1016/j.jsxm.2017.09.004.
34. Maathz P, McCracken LM, Eriksson V, Säde F, Aneblom G, Rikner Å, et al. A feasibility trial of online acceptance and commitment therapy for women with provoked vestibulodynia. Scand J Pain. 2023;23(3):476–82. https://doi.org/10.1515/sjpain-2022-0146.
35. Masheb RM, Kerns RD, Lozano C, Minkin MJ, Richman S. A randomized clinical trial for women with vulvodynia: Cognitive-behavioral therapy vs. supportive psychotherapy. Pain. 2009;141(1–2):31–40. https://doi.org/10.1016/j.pain.2008.09.031.
36. Morin M, Dumoulin C, Bergeron S, Mayrand MH, Khalifé S, Waddell G, et al. Multimodal physical therapy versus topical lidocaine for provoked vestibulodynia: a multicenter, randomized trial. Am J Obstet Gynecol. 2021;224(2):189.e1-189.e12. https://doi.org/10.1016/j.ajog.2020.08.038.
37. Murina F, Bianco V, Radici G, Felice R, Di Martino M, Nicolini U. Transcutaneous electrical nerve stimulation to treat vestibulodynia: a randomised controlled trial. BJOG. 2008;115(9):1165–70. https://doi.org/10.1111/j.1471-0528.2008.01803.x.
38. Murina F, Felice R, Di Francesco S, Oneda S. Vaginal diazepam plus transcutaneous electrical nerve stimulation to treat vestibulodynia: a randomized controlled trial. Eur J Obstet Gynecol Reprod Biol. 2018;228:148–53. https://doi.org/10.1016/j.ejogrb.2018.06.026.
39. Murina F, Graziottin A, Felice R, Radici G, Tognocchi C. Vestibulodynia: synergy between palmitoylethanolamide + transpolydatin and transcutaneous electrical nerve stimulation. J Low Genit Tract Dis. 2013;17(2):111–6. https://doi.org/10.1097/LGT.0b013e3182652316.
40. Nyirjesy P, Sobel JD, Weitz MV, Leaman DJ, Small MJ, Gelone SP. Cromolyn cream for recalcitrant idiopathic vulvar vestibulitis: results of a placebo controlled study. Sex Transm Infect. 2001;77(1):53–7. https://doi.org/10.1136/sti.77.1.53.
41. Petersen CD, Giraldi A, Lundvall L, Kristensen E. Botulinum toxin type A-a novel treatment for provoked vestibulodynia? Results from a randomized, placebo controlled, double blinded study. J Sex Med. 2009;6(9):2523–37. https://doi.org/10.1111/j.1743-6109.2009.01378.x.
42. Hullender Rubin LE, Mist SD, Schnyer RN, Chao MT, Leclair CM. Acupuncture augmentation of lidocaine for provoked, localized vulvodynia: a feasibility and acceptability study. J Low Genit Tract Dis. 2019;23(4):279–86. https://doi.org/10.1097/LGT.0000000000000489.
43. Schlaeger JM, Xu N, Mejta CL, Park CG, Wilkie DJ. Acupuncture for the treatment of vulvodynia: a randomized wait-list controlled pilot study. J Sex Med. 2015;12(4):1019–27. https://doi.org/10.1111/jsm.12830.
44. Merlino L, Ciminello E, Volpicelli AI, Tillier S, Pasquali MF, Dominoni M, et al. Evaluation of the effectiveness of combined treatment with intravaginal diazepam and pelvic floor rehabilitation in patients with vulvodynia by ultrasound monitoring of biometric parameters of pelvic muscles: a pilot study. Diseases. 2024;12(8):174. https://doi.org/10.3390/diseases12080174.
45. Bergeron S, Binik YM, Khalifé S, Pagidas K, Glazer HI, Meana M, et al. A randomized comparison of group cognitive--behavioral therapy, surface electromyographic biofeedback, and vestibulectomy in the treatment of dyspareunia resulting from vulvar vestibulitis. Pain. 2001. https://doi.org/10.1016/S0304-3959(00)00449-8.
46. De Seta F, Stabile G, Antoci G, Zito G, Nappi RE. Provoked vestibulodynia and topical treatment: a new option. Healthcare (Basel). 2022;10(5):830. https://doi.org/10.3390/healthcare10050830.
47. Farajun Y, Zarfati D, Abramov L, Livoff A, Bornstein J. Enoxaparin treatment for vulvodynia: a randomized controlled trial. Obstet Gynecol. 2012;120(3):565–72. https://doi.org/10.1097/AOG.0b013e3182657de6.
48. Morin A, Léonard G, Gougeon V, Cyr MP, Waddell G, Bureau YA, et al. Efficacy of transcranial direct-current stimulation in women with provoked vestibulodynia. Am J Obstet Gynecol. 2017;216(6):584.e1-584.e11. https://doi.org/10.1016/j.ajog.2017.02.049.
49. Murina F, Recalcati D, Di Francesco S, Cetin I. Effectiveness of two transcutaneous electrical nerve stimulation (TENS) protocols in women with provoked vestibulodynia: a randomized controlled trial. Med Sci (Basel). 2023;11(3):48. https://doi.org/10.3390/medsci11030048.
50. Moravek MB, Legocki LJ, Piper CK, Bernard K, Reed BD, Haefner HK. Impact of a single-session psychosocial counseling intervention for women with vulvodynia. Int J Gynaecol Obstet. 2023;160(1):202–8. https://doi.org/10.1002/ijgo.14321.
51. Higgins JP, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD, et al. The cochrane collaboration's tool for assessing risk of bias in randomised trials. BMJ. 2011;343:d5928. https://doi.org/10.1136/bmj.d5928.
52. Cohen P, Cohen J, Aiken LS, West SG. The problem of units and the circumstance for POMP. Multivar Behav Res. 1999;34(3):315–46. https://doi.org/10.1207/S15327906MBR3403_2.
53. Hopkins WG, Rowlands DS. Standardization and other approaches to meta-analyze differences in means. Stat Med. 2024;43(16):3092–108. https://doi.org/10.1002/sim.10114.
54. Cuijpers P, Weitz E, Cristea IA, Twisk J. Pre-post effect sizes should be avoided in meta-analyses. Epidemiol Psychiatr Sci. 2017;26(4):364–8. https://doi.org/10.1017/S2045796016000809.
55. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.5 (updated August 2024). Cochrane, 2024. Available from www.cochrane.org/handbook.
56. Béliveau A, Boyne DJ, Slater J, Brenner D, Arora P. BUGSnet: an R package to facilitate the conduct and reporting of Bayesian network meta-analyses. BMC Med Res Methodol. 2019;19(1):196. https://doi.org/10.1186/s12874-019-0829-2.
57. Owen RK, Bradbury N, Xin Y, Cooper N, Sutton A. MetaInsight: an interactive web-based tool for analyzing, interrogating, and visualizing network meta-analyses using R-shiny and netmeta. Res Synth Methods. 2019;10(4):569–81. https://doi.org/10.1002/jrsm.1373.
58. Spiegelhalter DJ, Best NG, Carlin BP, Van Der Linde A. Bayesian measures of model complexity and fit. J R Stat Soc Ser B Stat Methodol. 2002;64(4):583–639. https://doi.org/10.1111/1467-9868.00353.
59. Lin L, Zhang J, Hodges JS, Chu H. Performing arm-based network meta-analysis in R with the pcnetmeta package. J Stat Softw. 2017;80:5. https://doi.org/10.18637/jss.v080.i05.
60. Nikolakopoulou A, Higgins JPT, Papakonstantinou T, Chaimani A, Del Giovane C, Egger M, et al. CINeMA: an approach for assessing confidence in the results of a network meta-analysis. PLoS Med. 2020;17(4):e1003082. https://doi.org/10.1371/journal.pmed.1003082.
61. Hutton B, Salanti G, Caldwell DM, Chaimani A, Schmid CH, Cameron C, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. Ann Intern Med. 2015;162(11):777–84. https://doi.org/10.7326/M14-2385.
62. Rosen NO, Dawson SJ, Brooks M, Kellogg-Spadt S. Treatment of vulvodynia: pharmacological and non-pharmacological approaches. Drugs. 2019;79(5):483–93. https://doi.org/10.1007/s40265-019-01085-1.
63. Calafiore D, Marotta N, Curci C, Agostini F, De Socio RI, Inzitari MT, et al. Efficacy of rehabilitative techniques on pain relief in patients with vulvodynia: a systematic review and meta-analysis. Phys Ther. 2024;104(7):pzae054. https://doi.org/10.1093/ptj/pzae054.
64. Queiroz JF, Sarmento ACA, Aquino ACQ, de Souza ATB, de Meiros KS, Falsetta ML, et al. Psychotherapy and psychotherapeutic techniques for the treatment of vulvodynia: a systematic review and meta-analysis. J Low Genit Tract Dis. 2025;29(3):280–7. https://doi.org/10.1097/LGT.0000000000000881.

65. Pérez-López FR, Bueno-Notivol J, Hernandez AV, Vieira-Baptista P, Preti M, Bornstein J. Systematic review and meta-analysis of the effects of treatment modalities for vestibulodynia in women. Eur J Contracept Reprod Health Care. 2019;24(5):337–46. https://doi.org/10.1080/13625187.2019.1643835.
66. Nascimento RP, Falsetta ML, Maurer T, Sarmento ACA, Gonçalves AK. Efficacy of physiotherapy for treating vulvodynia: a systematic review. J Low Genit Tract Dis. 2024;28(1):54–63. https://doi.org/10.1097/LGT.0000000000000787.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.